

G-03-local-epsilon-regularity v1: audit exact local theorem for C_G

ZFL Lab

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MISSION: Determine whether external local epsilon-regularity yields exact C_G from G-01.
 TARGET G-01 $C_G^{L^3}$: $\exists t_n \uparrow T_*, x_n, r_n \downarrow 0 : \|u(t_n)\|_{L^3(B(x_n, r_n))} \geq \epsilon_0$

FORBIDDEN: $L_t^\infty L^3(B_1) \Rightarrow$ regularity without theorem match. CKN citation as substitute for exact implication. ESS-global $\Rightarrow$ local epsilon-regularity.

AUDIT:

1. Exact theorem search:

Candidate A: CKN ϵ -regularity (1982): Suitable weak (u, p) in $Q_1 := B_1 \times (-1, 0)$, $\mathcal{Q}(r) := r^{-2} \int_{Q_r} |u|^3 + r^{-2} \int_{Q_r} |p|^{3/2}$. If $\mathcal{Q}(1) \leq \epsilon_{CKN}$ universal, then regular in $Q_{1/2}$. Quantity is space-time + pressure, not instantaneous $L^3(B)$.

Candidate B: Kozono-Taniuchi / ESS local L^3 : If $\limsup_{r \rightarrow 0} r^{-3} \|u\|_{L_t^\infty L^3(Q_r)}^3$ small? Formulation varies. Most statements require $u \in L_t^\infty L_x^3(Q_1)$ and smallness of $\|u\|_{L_t^\infty L^3}$, plus suitable.

Candidate C: Seregin's L^3 epsilon-regularity: $\|u\|_{L_t^\infty L^3(Q_1)} \leq \epsilon_*$ implies regularity, but needs suitable + pressure bound from local energy inequality.

Issue: No theorem found so far of pure form $\|u(t)\|_{L^3(B_1)} \leq \epsilon_* \Rightarrow$ regularity at time t without space-time and pressure.

2. Suitable-weak hypotheses: All CKN-type require $u \in L_t^\infty L^2 \cap L_t^2 H_{loc}^1$, $p \in L_{loc}^{3/2}$, local energy inequality.

3. Space-time vs instantaneous: CKN quantity $\mathcal{Q}$ involves $\int_{-r^2}^0 \int_{B_r} |u|^3$, not $\|u(t_n)\|_{L^3(B_{r_n})}$ at single time. So contrapositive gives: If singular at (x_*, T_*) , then $\limsup_{r \downarrow 0} \mathcal{Q}(r) \geq \epsilon_{CKN}$. This is C_G^{CKN} , not $C_G^{L^3}$.

4. Scaling: $\mathcal{Q}(r)$ scale invariant: $|u|^3 \sim r^{-3}$, $dxdt \sim r^5$, so $r^{-2} \int |u|^3 \sim 1$. Matches L^3 critical. $\|u(t)\|_{L^3(B_r)}$ also critical: scaling $r^{-1} \|u\|_3^3$.

5. Pressure: C_G^{CKN} includes p . Cannot drop without theorem. $C_G^{L^3}$ would require pressure controlled by u via Riesz, not immediate locally.

6. Universality ϵ_* : CKN ϵ_{CKN} universal (depends only on dimension). If theorem valid, universal.

7. Does it give regularity at required point/scale? CKN gives: if singular at (x_*, T_*) , then $\mathcal{Q}(r) \geq \epsilon_*$ for all small r (up to rescaling). So produces $r_n \downarrow 0$.

8. Exact contrapositive producing C_G :

If $T_* < \infty$ and (x_*, T_*) singular point, then for $Q_{r_n}(x_*, T_*)$,

$$\mathcal{Q}(r_n) = r_n^{-2} \int_{Q_{r_n}} |u|^3 + r_n^{-2} \int_{Q_{r_n}} |p|^{3/2} \geq \epsilon_{CKN}$$

Call this C_G^{CKN} .

Thus: $C_G^{L^3} \neq C_G^{CKN}$ automatically.

$C_G^{L^3}$ NOT DERIVED from CKN alone

C_G^{CKN} PROVEN CONDITIONAL on CKN epsilon-regularity as external dependency
STATUS:

$C_G^{L^3}$ remains OPEN/BLOCKED: needs local $L_t^\infty L^3$ theorem exact

C_G^{CKN} PROVEN-CONDITIONAL: EXTERNAL-DEPENDENCY: CKN-1982

Hence split:

$C_G^{L^3}$: G-01 candidate, OPEN.

C_G^{CKN} : new candidate, PROVEN-CONDITIONAL on CKN.

$\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$ untouched. C_G layer separated.

Next CROSS should use C_G^{CKN} explicitly, not conflate.